

Eye Movement Analysis of Human Visual Recognition Processes with Camera Eye Tracker: Higher Mean and Variance of Fixation Duration for Familiar Images

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ABSTRACT

Eye movements could provide information on a person's cognitive context. Previous works showed that human visual recognition processes could be detected via electrooculography (EOG) data and revealed top-ranked eye movement features for detection. However, they lack analysis on how eye movements differ when viewing recognised (familiar) and unrecognised (unfamiliar) images. EOG methods are less prevalent and have limited scalability for detecting other cognitive processes. Our study with 34 participants showed that saccade count and fixation count decreased while the mean and variance of fixation duration increased with image familiarity. We discuss the differences in eye movements when seeing images of different categories and lessons learnt for our next steps to enable implicit interactions using wearable camera-based eye trackers.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; User studies.

KEYWORDS

Eye Movement Analysis, Cognitive Context, Visual Memory Recall, Visual Recognition, Eye Tracking, Image Familiarity, Physiological Sensing

ACM Reference Format:

Samantha Chan, Haimo Zhang, and Suranga Nanayakkara. 2023. Eye Movement Analysis of Human Visual Recognition Processes with Camera Eye Tracker: Higher Mean and Variance of Fixation Duration for Familiar Images. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3544549.3585782>

1 INTRODUCTION

Eye movements can provide rich information related to our cognitive context and enable understanding of our cognitive processes [34]. Our visual recognition of people, places and objects around us is essential for everyday life. Visual recognition processes can be detected using feature-based classification through electrooculogram

(EOG) eye tracking to infer when a user sees a new (unrecognised and unfamiliar) image and a repeated (recognised and familiar) image [4, 24]. Among the use cases for such a system, it could be useful when integrated with a memory assistant that recognises when you have forgotten a person's name [4] or details about a place and provides information about the person or place when implicitly needed. Previous works indicate highly-relevant eye movement features and parameters for the detection of visual recognition [4, 5, 24]. However, they lack a deeper understanding and analysis of the nature of the differences in eye movements between viewing recognised and unrecognised images. For example, saccade feature groups are important features for EOG-based visual recognition detection, it remains unknown if parameters like saccade count increase or decrease with familiarity with the contents of the image. The EOG method allows for portable and wearable eye tracking but is not prevalent in existing devices and has limited scalability for detecting other kinds of cognitive processes, such as cognitive load via pupil size.

This paper presents our research-in-progress where we conducted a study ($n=34$) to investigate how eye movement behaviours change when users recognise or do not recognise an image and if it was feasible to use camera-based eye movement trackers to detect visual recognition. This paper contributes with an eye movement analysis that provides a better understanding and insights into our eye movement patterns during visual recognition processes. Our findings reveal that saccade count and fixation count decrease with image familiarity while the mean and variance of fixation duration increase with image familiarity. We discuss the differences in eye movements when seeing images of various categories. Our results show that it is feasible to attain significant differences in parameters shown in prior works to be highly relevant to visual recognition. This work is a building block for our next research phase with the goal of enabling implicit interactions based on varying cognitive contexts using wearable camera-based eye trackers.

2 RELATED WORK

This work was inspired by prior studies on detecting cognitive contexts and visual recognition processes, especially through eye movements. There have been many works showing the connection between our visual, cognitive and memory systems. Eye-tracking methods can be used to identify if a user recognises an image [4, 24]. Studies showed that our gaze diverts to facilitate remembering [14, 20, 31]. Using eye movements, we can detect the cognitive context of mind wandering during reading [3]. Eye vergence behaviour, which

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CHI EA '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585782>

relates to the distance between the gaze points of each eye, has been used to detect internal thought [18]. We see that a combination of gaze patterns, eye movements and biosignals can be used to detect when older adults engage in visual search for items in the real-world [11]. Gaze was shown to be an attention indicator in the aided eyes just-in-time memory system [19]. The cognitive contexts of mental visualising and recalling a mental image can be detected through spatio-temporal mapping of eye movements [32]. The onset of working memory encoding can be detected at an above-chance success rate in realistic virtual reality (VR) environments [27]. Eye movements can also be used for everyday activity recognition [29] and to recognise document types during reading [22]. Lastly, eye movements have been shown to allow systems to predict intentions, especially intended actions such as picking up a cup [1, 26]. These works show the range of cognitive contexts that could be sensed through eye-related movements and responses.

In this paper, we mainly build on the work from a prior study that used EOG electrode sensors for eye-tracking and feature-based machine learning classification to detect *visual recognition processes* (whether one recognises an image or not) [4]. EOG eye-tracking is usually portable and wearable with electrodes placed on the face around the user's eyes. The study revealed that the highest-ranking eye movement features for detecting when users view familiar (recognised) or unfamiliar (unrecognised) images are the *variance of the fixation duration* and *saccade features* [4]. As a face or image becomes more familiar (with increasing number of repeated exposures to the image), the mean fixation count has been shown to decrease [5, 17]. Apart from another work that detected visual recognition via EOG signals for visual-auditory stimuli, there are no other systems to date to detect visual recognition processes.

Our work is different from the prior works in two ways. First, the main prior work [4] used feature-based machine learning and feature ranking to provide insights on the types of eye movement features that are useful for assessing visual memory recognition processes. This work expands these insights to evaluate the differences in eye movement features when seeing repeated (recognised) and new (unrecognised) images. For example, the variance of fixation duration is a useful feature proposed in the prior study. We intend to *understand* if the variance is higher or lower when viewing new images compared to repeated images. Second, we use camera-based eye-tracking due to its prevalence and adoption in commercial products such as wearable¹ and augmented reality (AR)/VR systems², laptops and desktop computer setups³. Unlike EOG methods, camera-based tracking enables the measurement of a larger range of eye-related parameters that we can infer cognitive states from. In particular, pupil diameters and responses cannot be measured with EOG but are well-known indexes of cognitive load [12, 30] and intention-related processing in event-based prospective memory tasks [25]. We argue that camera-based eye-tracking is important as our research goal is to develop a scalable system with a more prevalent tracking method and one that can eventually be used to detect other cognitive states.

¹<https://pupil-labs.com/products/invisible/>

²<https://vr.tobii.com/integrations/htc-vive-pro-eye/>

³<https://gaming.tobii.com/products/laptops/>

3 STUDY

The study used a within-subjects design and investigated eye movements during visual recognition processes. The study goal was to record and compare the eye movements for two conditions: 1) when viewing *new* images and 2) when viewing *repeated* images. Based on previous studies [4, 17, 24], we assume that the *new* images are unrecognised and unfamiliar while the *repeated* images are recognised and familiar.

3.1 Participants

Participants were recruited by email and word-of-mouth. They were included in the study if they had normal or corrected-to-normal eyesight and were fluent or native English speakers. Thirty-four participants took part in the study (mean age = 43 years, age range = 18 to 72, 18 male and 16 female). Ethical approval was obtained prior to the study and participants gave written consent to use their information.

3.2 Apparatus and Stimuli

The experiment stimuli were images shown using MATLAB 2019a on a screen with a resolution of 1920 x 1200. Participants were seated approximately 60 cm from the screen (Figures 1a and b). A Tobii EyeX infra-red camera eye-tracker, which was mounted on the bottom edge of the screen, was used to record eye movements and had a sampling frequency of 50 Hz.

Following stimuli selection by prior works [4, 5], the images were categorised into four themes: *abstract*, *landscape* (lakes), *buildings* (skyscrapers), and *faces* (frontal view). Face images were taken from the CVL face database [28] and the rest were chosen and taken from online image searches (see Appendix A). All participants were presented with the same set of images.

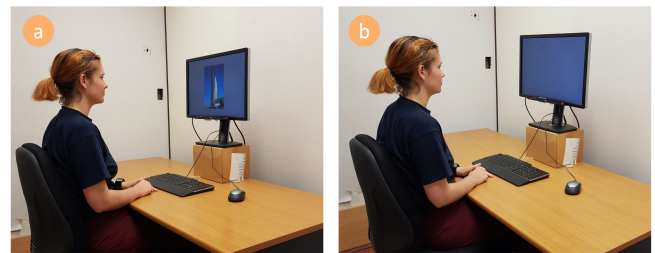


Figure 1: Participant is seated in front of screen and desktop eye-tracker. The screen displayed (a) the image stimuli and (b) a blank gaussian noise screen during the baseline and breaks.

3.3 Procedure

Participants were told that their eye movements would be recorded while they saw four sets of images and were asked to sit and simply view the images. They were not informed of the experimental conditions nor about the repeated images. Participants were presented with nine images for each of the four themes. Each of the nine images was shown once and three randomly-selected images were shown again for three more times. The order of image presentation

was randomised. Thus, nine data segments were collected for each condition (9 *new* and 9 *repeated*). Each image was shown for five seconds (for example, in Figure 1a). There was a three-second break between each image in which a blank grey screen was shown (Figure 1b). The experiment duration was about 15 minutes to minimise fatigue in participants [4, 24]. At the end of the study, they were asked to confirm if they had seen any of the displayed images before the study.

3.4 Analysis

Each data segment recorded the timestamp, x and y gaze positions in pixels. We conducted eye movement analysis for each data segment using a modified version [7] of PyTrack [13] and PyGaze Analyzer [10]. The modifications centred on calculating eye movement parameters/features that were similar to those used in the prior studies on detecting visual recognition. Our analysis first detected eye movements of fixation, saccades and blinks. Fixation refers to a period of time of 100 milliseconds (ms) or more in which the person's gaze is fixed within a small region. A saccade usually refers to the quick movements between fixations. A blink is defined as eyelid closure with a duration of 50 to 500 ms. Parameters, such as count and duration, were extracted from the detected fixations, saccades and blinks. Although we extracted 39 parameters (see Appendix B) that were suggested in prior studies [4, 5], we focused our report on parameters that have been shown in related works as highly-related to detecting visual recognition. These parameters were the mean saccade count [4], the mean fixation count [5, 17], the mean fixation duration and the variance of fixation duration [4, 24]. For our analysis of each image theme, we report additional parameters that are meant to supplement the analysis and discussion of the results. Since Shapiro-Wilk Tests showed that the parameters were normally distributed ($p > .05$), paired t-tests were conducted to compare the differences in the mean values of the extracted parameters between each condition.

4 RESULTS

For each person, 72 data segments were collected (36 new and 36 repeated). This resulted in 2448 data segments (1224 per condition). In this section, we present our analysis results of data segments for (1) all images and then for each image theme: (2) abstract images, (3) images of buildings, (4) images of faces, and (5) images of landscapes. Participants confirmed that they had not seen any of the images prior to the study.

4.1 All Images

Participants had significantly lower saccade count ($p < .001$, Figure 2a) and fixation count ($p < .001$, Figure 2b) when viewing the repeated images compared to viewing the new images. The mean fixation duration ($p < .01$, Figure 2c) and variance of fixation duration ($p < .05$, Figure 2d) were significantly higher when viewing repeated images than when viewing new images. Statistical values for the tests are summarised in Table 1.

4.2 Abstract Images

Participants had significantly lower saccade counts when viewing repeated abstract images than new ones ($p < .05$, Figure 3a). There

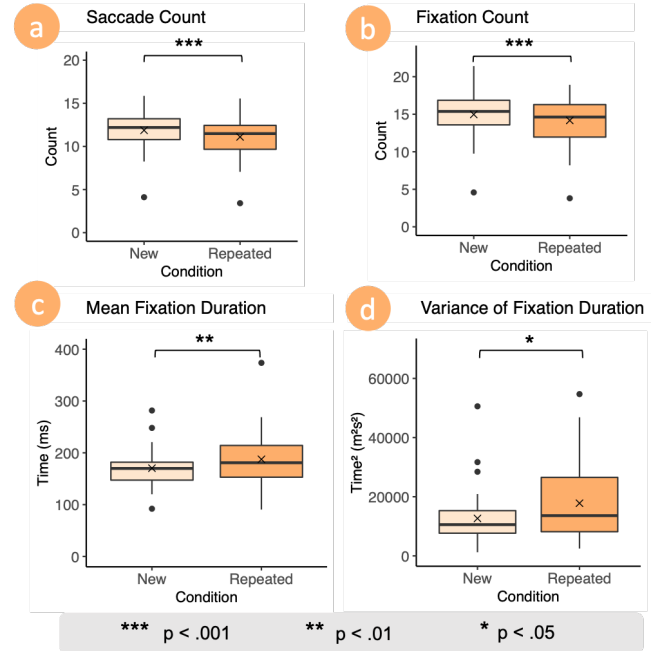


Figure 2: Boxplots for (a) saccade count, (b) fixation count, (c) mean fixation duration and (d) variance of fixation duration for all images per condition.

were no significant differences in the fixation count, mean of fixation duration and variance of fixation duration between viewing repeated and new abstract images. Further analysis shows that the variance ($p < .05$) of saccade length in the horizontal (x) direction was significantly lower for repeated abstract images than new abstract images (refer to summary of values in Table 2).

4.3 Images of Buildings

There was a significantly lower saccade count ($p < .001$, Figure 3b) and fixation count ($p < .001$) when viewing repeated images of buildings than new ones. Participants had significantly higher mean fixation duration ($p < .01$) when viewing repeated images than new ones ($p < .001$). We found no significant differences in the variance of fixation duration between repeated and new conditions. Our supplementary analysis reveals that participants' mean saccade duration ($p < .05$) and maximum horizontal (x) saccade length ($p < .05$) were significantly higher when seeing repeated images compared to new images. Their maximum blink duration ($p < .01$) was significantly lower for new images versus repeated images.

4.4 Images of Faces

Participants had significantly lower saccade count ($p < .01$, Figure 3c) and fixation count ($p < .001$) for repeated images of faces compared to new images. Their mean fixation duration ($p < .01$), variance of fixation duration ($p < .01$) and maximum blink duration ($p < .01$) were all significantly higher when viewing repeated images than when viewing new images.

Summary of Values for All Images				
Parameters	New Images ($M \pm SD$)	Repeated Images ($M \pm SD$)	t_{33} Value	p-Value
Saccade Count	11.9 $\pm$ 2.23	11.1 $\pm$ 2.38	5.45	<.001
Fixation Count	15.0 $\pm$ 3.09	14.2 $\pm$ 3.17	4.26	<.001
Mean Fixation Duration	170 $\pm$ 37.4	187 $\pm$ 52.7	-3.40	<.01
Var. Fixation Duration	12666 $\pm$ 9418	25242 $\pm$ 34776	-2.34	<.05

Table 1: Summary of mean values (M), standard deviations (SD), t-values and p-values for the parameters for all images.

4.5 Images of Landscapes

Lastly, there was a significantly lower saccade count when participants viewed repeated images compared to new ones ($p < .05$, Figure 3d). We did not find any significant differences in the fixation count, mean of fixation duration or variance of fixation duration between the conditions. Additional analysis shows that the variance ($p < .01$) of vertical velocity for repeated images was significantly higher than for repeated images.

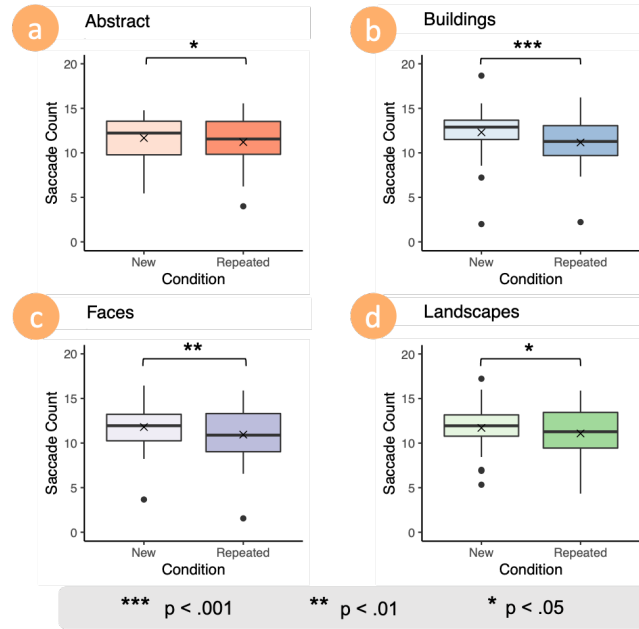


Figure 3: Boxplots of saccade count per condition for the image themes: (a) abstract, (b) buildings, (c) faces, and (d) landscapes.

5 DISCUSSION

The results show that saccade count was a key parameter in all images and each image theme. As the images are repeated and become more familiar, the saccade count decreases. This may mean that less scanning is done for familiar images, which seems to correspond to the previous findings that we have more efficient scanning for familiar faces [17]. Our findings that there were fewer fixations when viewing repeated images aligns with prior findings [5, 17] that fixation count decreases with familiarity and repeated exposure

to the images. In another sense, we tend to fixate on the various elements within the images when seeing an image for the first time, probably to analyse or remember different parts of the image. This aligns with knowledge that observers tend to view the outlines and contours of visual images when seeing them for the first time and when seeing the images continuously or subsequently, observers tended not to view the borders and contours again [33]. We found that the mean fixation duration and variance of fixation duration are higher when viewing familiar images. Our finding adds insight and understanding to why the mean and variance of fixation duration were highly ranked as useful features for visual recognition detection models in previous works [4]. This could indicate that when viewing familiar images, people tend not to fixate on various parts of the image as they do on new images but instead fixate longer on a few parts of the image.

When we examine the results in terms of the image categories, lowered fixation count is only shown for familiar buildings and faces while there were no significant differences in fixation count when viewing repeated and new landscape and abstract images. This observation might be because buildings and faces have salient image features to focus on [33], such as the shape of the building or facial features. On the other hand, abstract images might not have specific salient features and landscapes have multiple things to focus on (e.g., trees, lakes, and sky). It also seems that the maximum blink duration was higher, indicating that people blink for longer durations when viewing repeated images of buildings and faces. Higher blink duration is an indicator of higher mental workload [2]. The lower variance of saccade length in the horizontal directions for repeated abstract images than new abstract images could suggest we tend to horizontally scan new abstract images. The higher maximum horizontal saccade length when viewing repeated images could tell us that we might start scanning more to the left and right around the buildings when seeing familiar images of buildings. For images of landscapes, there was a higher variation in the vertical saccade velocity for familiar images. This might mean we have quick scanning patterns up and down for landscape images, perhaps switching between visual elements in the background and foreground.

5.1 Summary of Implications

Our findings and discussion present a deeper understanding and insights into our eye movement behaviours during visual recognition processes, particularly when viewing familiar and unfamiliar images. We also discussed the potential differences in viewing patterns for images of various categories. The implications of this work are that it is feasible to attain significant differences in parameters

Summary of Values				
Parameters	New Images ($M \pm SD$)	Repeated Images ($M \pm SD$)	t_{33} Value	p-Value
Abstract				
Saccade Count	11.7 $\pm$ 2.30	11.2 $\pm$ 2.75	2.16	<.05
Fixation Count	14.7 $\pm$ 3.20	14.3 $\pm$ 3.78	1.28	n.s.
Mean Fixation Duration	168 $\pm$ 46.0	181 $\pm$ 62.1	-1.59	n.s.
Var. Fixation Duration	13334 $\pm$ 17500	18243 $\pm$ 34332	-.828	n.s.
Var. x Saccade Len.	77475 $\pm$ 89555	61943 $\pm$ 73970	2.13	<.05
Buildings				
Saccade Count	12.3 $\pm$ 2.86	11.2 $\pm$ 2.79	4.34	<.001
Fixation Count	15.4 $\pm$ 3.62	14.1 $\pm$ 3.54	4.27	<.001
Mean Fixation Duration	170 $\pm$ 39.7	190 $\pm$ 53.3	-2.83	<.01
Var. Fixation Duration	11703 $\pm$ 10065	29546 $\pm$ 56658	-1.86	n.s.
Max. Blink Duration	198 $\pm$ 156	233 $\pm$ 187	-2.88	<.01
Max. x Saccade Len.	395 $\pm$ 333	470 $\pm$ 445	-2.28	<.05
Faces				
Saccade Count	11.8 $\pm$ 2.65	10.9 $\pm$ 3.17	3.33	<.01
Fixation Count	15.5 $\pm$ 3.54	14.3 $\pm$ 3.99	3.71	<.001
Mean Fixation Duration	178 $\pm$ 45.8	201 $\pm$ 77.1	-3.14	<.01
Var. Fixation Duration	13806 $\pm$ 11323	27176 $\pm$ 30246	-3.12	<.01
Max. Blink Duration	235 $\pm$ 155	269 $\pm$ 229	-1.37	<.01
Landscapes				
Saccade Count	11.7 $\pm$ 2.50	11.1 $\pm$ 2.81	2.51	<.05
Fixation Count	14.3 $\pm$ 3.46	14.1 $\pm$ 3.68	1.09	n.s.
Mean Fixation Duration	165 $\pm$ 36.1	177 $\pm$ 63.4	-1.83	n.s.
Var. Fixation Duration	11820 $\pm$ 10495	26002 $\pm$ 79332	-1.13	n.s.
Var. y Saccade Vel.	5.71 $\pm$ 6.94	12.0 $\pm$ 12.2	-2.96	<.01

Table 2: Summary of mean values (M), standard deviations (SD), t-values and p-values for the parameters for image themes.

shown in prior works to have high relation to visual recognition, and camera-based eye trackers can be used for visual recognition detection. The findings in this paper are incremental yet required steps in our work-in-progress to build implicit interactions enabled by cognitive context sensing. Detection of visual recognition processes (as the cognitive context in this case) could enable interfaces to infer your familiarity with an image or a visual scene [4, 7]. If the system can infer your familiarity, for example, with a face, the system could then modulate the level of information given to you. Perhaps providing more information if you are not familiar and less information when you are. We believe it could be applied to support memory augmentation (e.g., via conversational interfaces [8, 9] and biosignal/physiological interfaces [6]), learning, and information retrieval in desktop, AR and VR settings [15, 16].

5.2 Lessons Learnt: Limitations and Future Work

The eye-tracker used in this study was an affordable, commercially available stationary desktop tracker. Our plan is to eventually transition into wearable or mobile eye-tracking to enable on-the-go cognitive context detection and support a broader range of activities and contexts. From our study, saccade and fixation detection and parameter extraction are essential for detecting visual recognition. The tracker's sampling frequency of 50 Hz may have reduced the capability of accurately detecting the eye movement parameters and attaining an equivalent amount of data points per participant

as in previous works [4, 24]. The previous studies used EOG tracker with high sampling rates of 128 Hz [4] and 256 Hz [24] which may be the cause of any discrepancies between those studies and this study. Since saccades involve quick eye movements that can be at a minimum duration of 5 milliseconds (ms) [10], 50 Hz might not allow us to detect as many saccades as there could be. The sampling rate of this study's tracker would only enable detection of saccades at a minimum of 20 ms ($1000\text{ms}/50 = 20\text{ms}$). Our next step is to use eye trackers with higher sampling frequency, such as the Pupil Core device by Pupil Labs [21], which is a wearable tracker that samples at 200 Hz.

We also plan to develop a detection model for visual recognition in the next phase of the research. While prior work shows that feature-based support vector machine (SVM) classification works for visual recognition for abstract images [4], we intend to build our inference model via graph-based methods [23].

Nonetheless, this paper shows we can get comparable results with prior studies even with a low sampling rate camera-based eye tracker. Our analysis enables better understanding of eye movement behaviours in visual recognition.

Our study used an existing face database and online images as stimuli. Although our participants verified that they did not see the stimuli images before the study, it might not be the same for other individuals. Future studies could use generative artificial intelligence (AI) methods to create entirely new stimuli images.

6 CONCLUSION

This work investigated how eye movements differ when users recognise or do not recognise an image and whether it was feasible to use camera-based eye movement trackers to detect visual recognition. Our study provided a better understanding of eye movement behaviour during visual recognition, showing that saccade count and fixation count decreased while the mean and variance of fixation duration increased with image familiarity. Camera-based eye trackers, including those with low sampling rates, can be used for visual recognition detection. Lessons learnt from this work will inform our ongoing work in developing implicit cognitive context aware interfaces, especially using wearable camera-based eye trackers.

ACKNOWLEDGMENTS

We would like to thank our participants and Jovana Lazarevic for their time and assistance. The face images from CVL Face Database used in this work have been provided by Computer Vision Laboratory, University of Ljubljana Slovenia [28]. This work was supported by Nanyang Technological University Singapore.

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A STIMULI IMAGES

A.1 Abstract

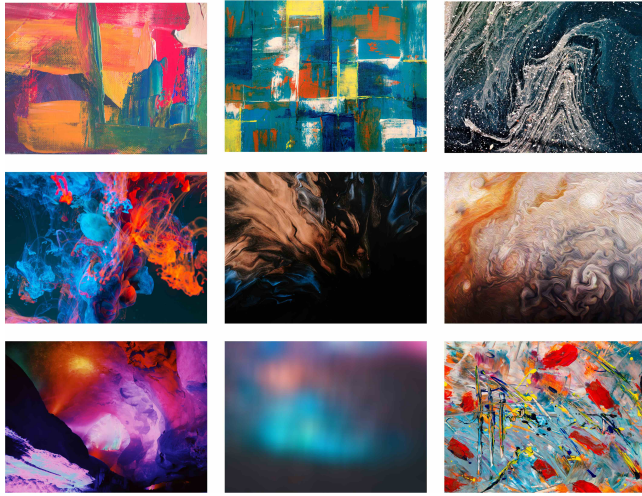


Figure 4: Selected images in abstract category

A.2 Landscape

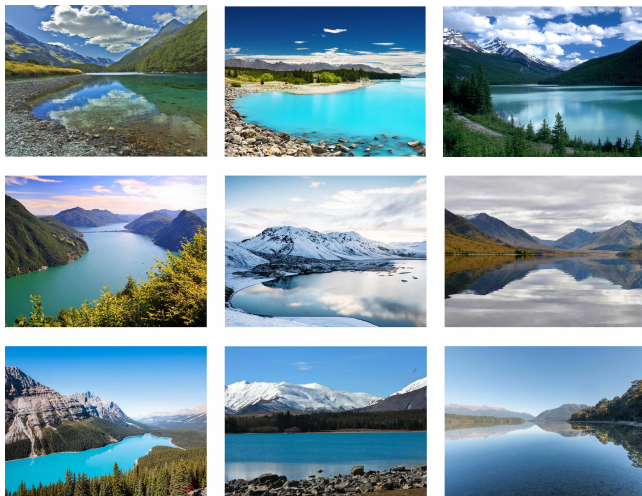


Figure 5: Selected images in landscape category

A.3 Buildings



Figure 6: Selected images in buildings category

A.4 Faces

Due to the CVL Face Database [28] licensing agreement, we are not allowed to redistribute the face images directly. We share the indexes to find the images on the database. We used the front-facing face images (MVC-003F) for database participants 1, 2, 4, 6, 10, 13, 18, 26, and 114.

B EYE MOVEMENT PARAMETERS

No.	Parameter	Description
1	FCount	Fixation Count
2	Fmax	Max. Fixation Count
3	Favg	Mean of Fixation Count
4	Fstd	SD of Fixation Count
5	Fvar	Var. of Fixation Count
6	BCount	Blink Count
7	Bmax	Max. Blink Count
8	Bavg	Mean of Blink Count
9	Bstd	SD of Blink Count
10	Bvar	Var. of Blink Count
11	SCount	Saccade Count
12	SmaxDur	Max. Saccade Duration
13	SavgDur	Mean of Saccade Duration
14	SstdDur	SD of Saccade Duration
15	SvarDur	Var. of Saccade Duration
16	SmaxVelx	Max. Saccade Velocity in x direction
17	SavgVelx	Mean of Saccade Velocity in x direction
18	SstdVelx	SD of Saccade Velocity in x direction
19	SvarVelx	Var. of Saccade Velocity in x direction
20	SmaxLenx	Max. Saccade Length in x direction
21	SavgLenx	Mean of Saccade Length in x direction
22	SstdLenx	SD of Saccade Length in x direction
23	SvarLenx	Var. of Saccade Length in x direction
24	SmaxVely	Max. Saccade Velocity in y direction
25	SavgVely	Mean of Saccade Velocity in y direction
26	SstdVely	SD of Saccade Velocity in y direction
27	SvarVely	Var. of Saccade Velocity in y direction
28	SmaxLeny	Max. Saccade Length in y direction
29	SavgLeny	Mean of Saccade Length in y direction
30	SstdLeny	SD of Saccade Length in y direction
31	SvarLeny	Var. of Saccade Length in y direction
32	SmaxVel	Max. Saccade Velocity
33	SavgVel	Mean of Saccade Velocity
34	SstdVel	SD of Saccade Velocity
35	SvarVel	Var. of Saccade Velocity
36	SmaxLen	Max. Saccade Length
37	SavgLen	Mean of Saccade Length
38	SstdLen	SD of Saccade Length
39	SvarLen	Var. of Saccade Length

Table 3: Extracted eye parameters for fixations, blinks and saccades.